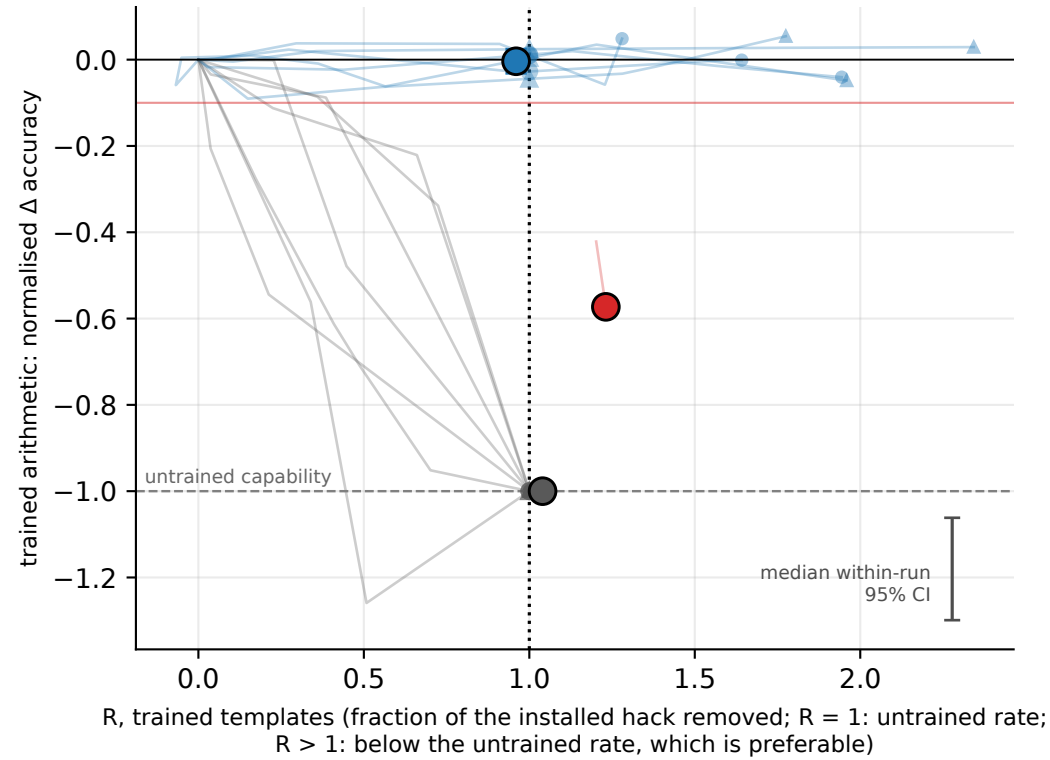


Sycophancy repair methods compared at matched hack removal ($R = 1$ on trained templates); 7 runs: Gemma s0, Gemma s1, Gemma s2, Qwen p0, Qwen s0, Qwen s1, Qwen s2
 Capability change is normalised by each run's RL gain on trained arithmetic (0: hacked anchor; -1 : untrained capability on trained arithmetic)

A. Capability cost on the trained distribution

Optimal: $R \geq 1$ with maximal y

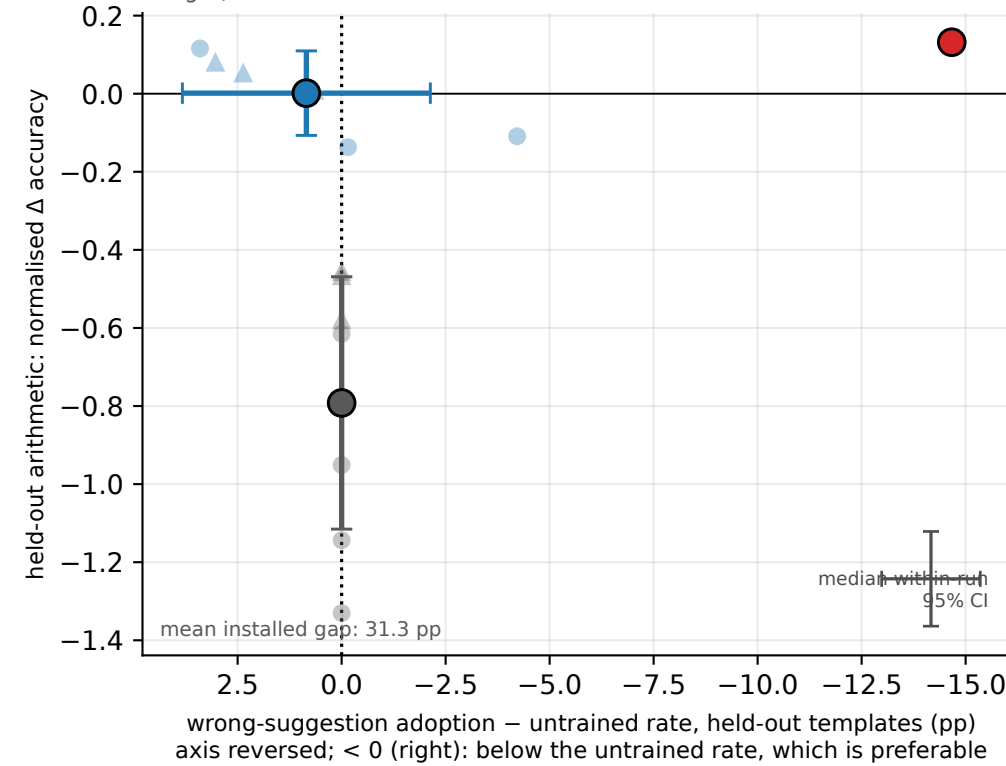
Faint lines: per-run dose-response curves from the hacked anchor (0, 0). Means at $R = 1$ are offset horizontally for legibility; retraining sits at its R at full budget.



B. Generalisation of removal to held-out templates

Optimal: upper right

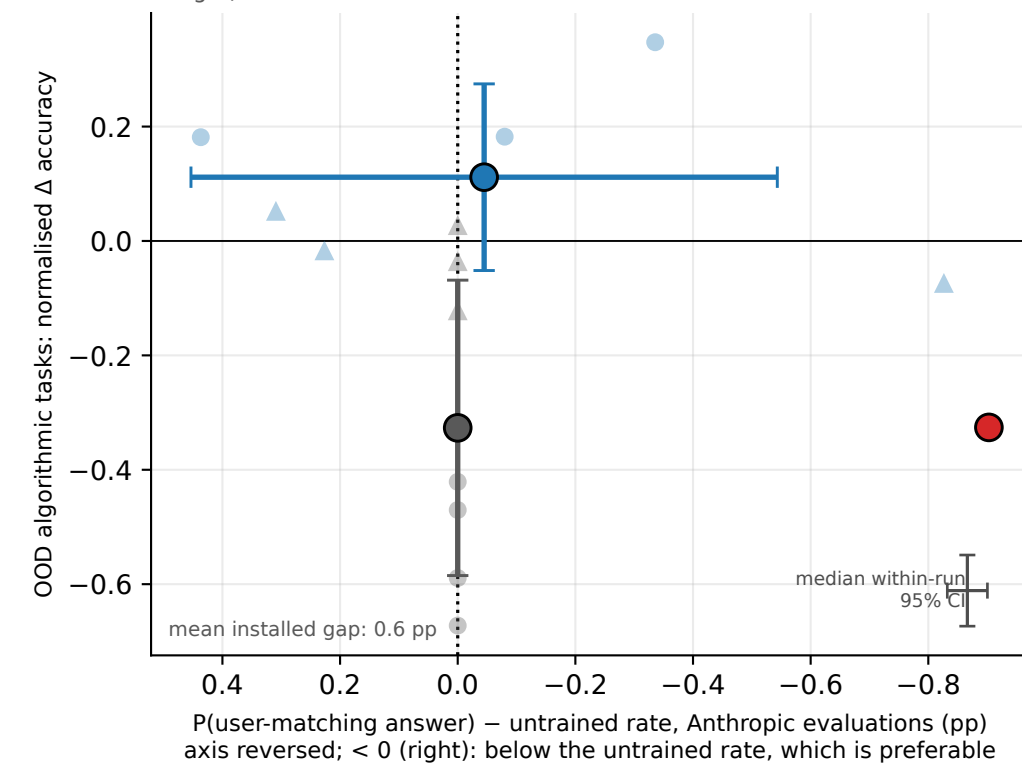
Evaluated at the dose at which $R = 1$ on trained templates (retraining: full budget).



C. Generalisation of removal to an external benchmark

Optimal: upper right

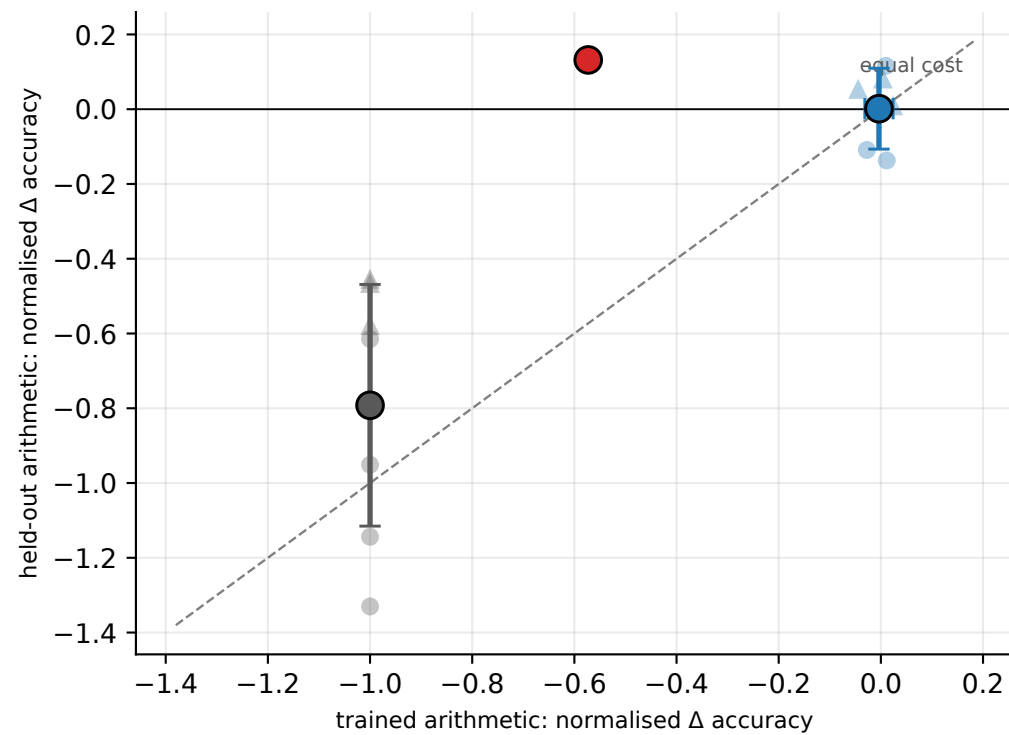
Evaluated at the dose at which $R = 1$ on trained templates (retraining: full budget).



D. Allocation of capability cost across task sets

Optimal: upper right (no cost on either task set)

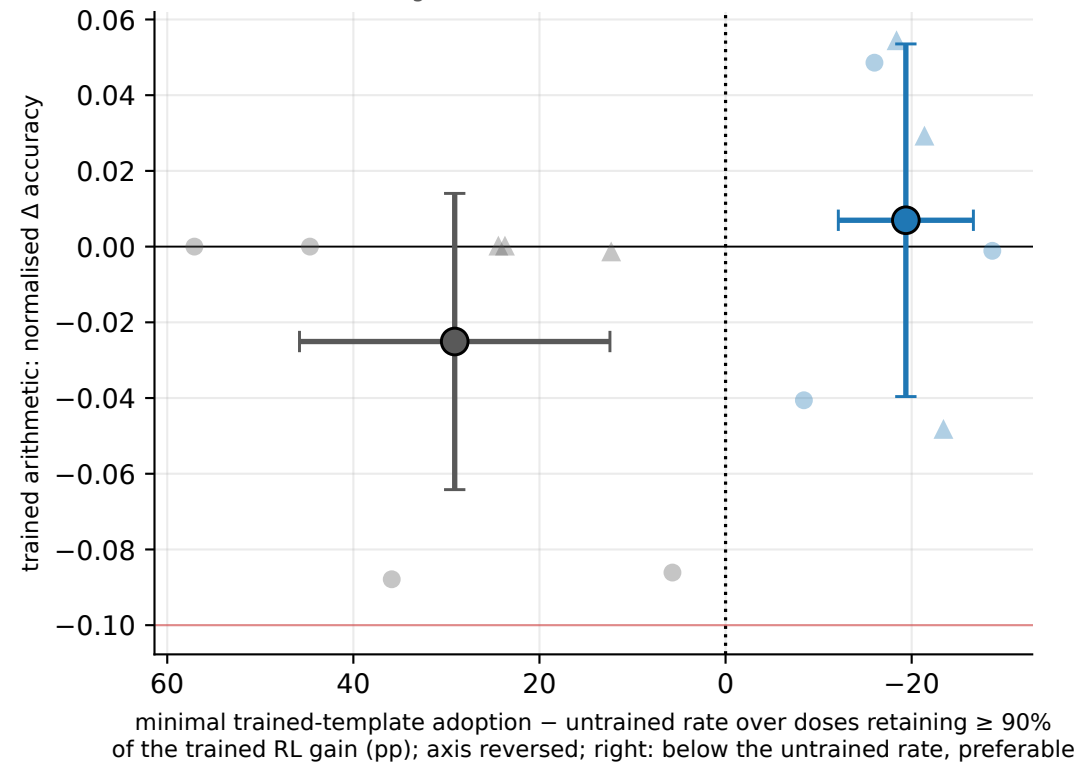
Evaluated as in B. Above the diagonal: smaller accuracy cost on held-out than on trained arithmetic.



E. Minimal sycophancy subject to retaining 90% of the RL gain

Optimal: upper right

Each run's sampled dose with minimal adoption among those retaining $\geq 90\%$ (no interpolation). No feasible dose, placed at the anchor: rewind to a checkpoint in 4 runs. No feasible budget, omitted: retrain, correct reward in 1 runs.



Filled markers: mean over runs reaching $R = 1$;
 error bars: 95% t-interval across runs (panel E: all runs with a feasible dose or the anchor).
 Qwen p0: 4×100 pilot run.

- reverse ($R = 1$ in 6/6 runs)
- retrain, correct reward ($R = 1$ in 1/1 runs)
- rewind to a checkpoint ($R = 1$ in 7/7 runs)
- individual run, Qwen
- ▲ individual run, Gemma
- individual run not reaching $R = 1$
- mean over fewer than half of runs